

Mathematics Section A

- Let  $f: \mathbf{R} \rightarrow \mathbf{R}$  be defined as  $f(x) = \frac{2x^2-3x+2}{3x^2+x+3}$ . Then  $f$  is :  
 (A) both one-one and onto (B) one-one but not onto  
 (C) onto but not one-one (D) neither one-one nor onto
- Consider the quadratic equation  $(n^2 - 2n + 2)x^2 - 3x + (n^2 - 2n + 2)^2 = 0, n \in \mathbf{R}$ . Let  $\alpha$  be the minimum value of the product of its roots and  $\beta$  be the maximum value of the sum of its roots. Then the sum of the first six terms of the G.P., whose first term is  $\alpha$  and the common ratio is  $\frac{\alpha}{\beta}$ , is :  
 (A)  $\frac{61}{37}$  (B)  $\frac{121}{81}$   
 (C)  $\frac{364}{243}$  (D)  $\frac{1093}{729}$
- Let  $S = \{z \in \mathbf{C} : z^2 + \sqrt{6}iz - 3 = 0\}$ . Then  $\sum_{z \in S} z^8$  is equal to:  
 (A) 162 (B) 184 (C) 262 (D) 324
- The sum of all possible values of  $\theta \in [0, 2\pi]$ , for which the system of equations :  

$$\begin{aligned} x \cos 3\theta - 8y - 12z &= 0 \\ x \cos 2\theta + 3y + 3z &= 0 \\ x + y + 3z &= 0 \end{aligned}$$
 has a non-trivial solution, is equal to:  
 (A)  $\pi$  (B)  $2\pi$  (C)  $3\pi$  (D)  $4\pi$
- Let  $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$  and  $B = [b_{ij}], 1 \leq i, j \leq 3$ . If  $B = A^{99} - I$ , then the value of  $\frac{b_{31}-b_{21}}{b_{32}}$  is :  
 (A) 99 (B) 199 (C) 149 (D) 159
- The sum  $1 + \frac{1}{2}(1^2 + 2^2) + \frac{1}{3}(1^2 + 2^2 + 3^2) + \dots$  upto 10 terms is equal to:  
 (A) 130 (B) 155  
 (C)  $\frac{315}{2}$  (D)  $\frac{325}{2}$
- A building has ground floor and 10 more floors. Nine persons enter in a lift at the ground floor. The lift goes up to the 10<sup>th</sup> floor. The number of ways, in which any 4 persons exit at a floor and the remaining 5 persons exit at a different floor, if the lift does not stop at the first and the second floors, is equal to :  
 (A) 2184 (B) 3064 (C) 7056 (D) 11340
- Let the mean and the variance of seven observations 2, 4,  $\alpha$ , 8,  $\beta$ , 12, 14,  $\alpha < \beta$ , be 8 and 16 respectively. Then the quadratic equation whose roots are  $3\alpha + 2$  and  $2\beta + 1$  is :  
 (A)  $x^2 - 35x + 306 = 0$  (B)  $x^2 - 41x + 420 = 0$   
 (C)  $x^2 - 45x + 506 = 0$  (D)  $x^2 - 37x + 342 = 0$
- A bag contains 6 blue and 6 green balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that each drawn pair consists of one blue and one green ball is :  
 (A)  $\frac{63}{925}$  (B)  $\frac{17}{231}$   
 (C)  $\frac{16}{231}$  (D)  $\frac{64}{925}$
- Let  $C$  be a circle having centre in the first quadrant and touching the  $x$ -axis at a distance of 3 units from the origin. If the circle  $C$  has an intercept of length  $6\sqrt{3}$  on the  $y$ -axis, then the length of the chord of the circle  $C$  on the line  $x - y = 3$  is :  
 (A) 8 (B) 6 (C)  $6\sqrt{2}$  (D)  $8\sqrt{2}$
- The eccentricity of an ellipse  $E$  with centre at the origin  $O$  is  $\frac{\sqrt{3}}{2}$  and its directrices are  $x = \pm \frac{4\sqrt{6}}{3}$ . Let  $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  be a hyperbola whose eccentricity is equal to the length of semi-major axis of  $E$ , and whose length of latus rectum is equal to the length of minor axis of  $E$ . Then the distance between the foci of  $H$  is :  
 (A)  $\frac{4\sqrt{2}}{\sqrt{7}}$  (B)  $\frac{4\sqrt{2}}{7}$   
 (C)  $\frac{4}{\sqrt{7}}$  (D)  $\frac{8}{7}$
- Let  $x = 9$  be a directrix of an ellipse  $E$ , whose centre is at the origin and eccentricity is  $\frac{1}{3}$ . Let  $P(\alpha, 0)$ ,  $\alpha > 0$ , be a focus of  $E$  and  $AB$  be a chord passing through  $P$ . Then the locus of the mid point of  $AB$  is :  
 (A)  $9y^2 = 8x(1 - x)$  (B)  $3y^2 = 4x(1 - x)$   
 (C)  $9y^2 = 8x(x - 1)$  (D)  $3y^2 = 4x(x - 1)$
- If  $\sin(\tan^{-1}(x\sqrt{2})) = \cot(\sin^{-1}\sqrt{1 - x^2})$ ,  $x \in (0, 1)$  then the value of  $x$  is :  
 (A)  $\frac{1}{2}$  (B)  $\frac{1}{3}$   
 (C)  $\frac{2}{3}$  (D)  $\frac{5}{8}$
- The shortest distance between the lines  $\frac{x-4}{1} = \frac{y-3}{2} = \frac{z}{-}$  and  $\frac{x+2}{2} = \frac{y-6}{4} = \frac{z-5}{-5}$  is:  
 (A)  $\frac{5\sqrt{6}}{6}$  (B)  $2\sqrt{5}$   
 (C)  $3\sqrt{5}$  (D)  $4\sqrt{5}$
- Let  $\vec{a} = 2\hat{i} + 3\hat{j} + 3\hat{k}$  and  $\vec{b} = 6\hat{i} + 3\hat{j} + 3\hat{k}$ . Then the square of the area of the triangle with adjacent sides determined by the vectors  $(2\vec{a} + 3\vec{b})$  and  $(\vec{a} - \vec{b})$  is:  
 (A) 450 (B) 900 (C) 1800 (D) 2400

16. Let  $\lim_{x \rightarrow 2} \frac{(\tan(x-2))(rx^2 + (p-2)x - 2p)}{(x-2)^2} = 5$  for some  $r, p \in \mathbf{R}$ . If the set of all possible values of  $q$ , such that the roots of the equation  $rx^2 - px + q = 0$  lie in  $(0, 2)$ , be the interval  $(\alpha, \beta]$ , then  $4(\alpha + \beta)$  equals :
- (A) 11 (B) 13 (C) 17 (D) 21
17. Let  $A = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 1 & \alpha \\ 0 & 1 & -1 \end{bmatrix}$  be a singular matrix. Let  $f(x) = \int_0^x (t^2 + 2t + 3) dt$ ,  $x \in [1, \alpha]$ . If  $M$  and  $m$  are respectively the maximum and the minimum values of  $f$  in  $[1, \alpha]$ , then  $3(M - m)$  is equal to :
- (A) 64 (B) 68 (C) 72 (D) 76
18. Let  $f: \mathbf{R} \rightarrow \mathbf{R}$  be such that  $f(xy) = f(x)f(y)$ , for all  $x, y \in \mathbf{R}$  and  $f(0) \neq 0$ . Let  $g: [1, \infty) \rightarrow \mathbf{R}$  be a differentiable function such that

$$x^2 g(x) = \int_1^x (t^2 f(t) - tg(t)) dt$$

Then  $g(2)$  is equal to :

- (A)  $\frac{13}{8}$  (B)  $\frac{11}{16}$   
(C)  $\frac{15}{32}$  (D)  $\frac{17}{64}$

19. The area of the region  $\{(x, y) : x^2 - 8x \leq y \leq -x\}$  is :
- (A)  $\frac{343}{6}$  (B)  $\frac{637}{6}$   
(C)  $\frac{437}{6}$  (D)  $\frac{523}{6}$
20. The value of the integral  $\int_{-1}^1 \left( \frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} \right) dx$  is equal to :
- (A)  $3 \log_e 2$  (B)  $2 \log_e 2$   
(C)  $5 \log_e 3$  (D)  $3 \log_e 3$

### Mathematics Section B

21. Let  $R = \{(x, y) \in \mathbf{N} \times \mathbf{N} : \log_e(x + y) \leq 2\}$ . Then the minimum number of elements, required to be added in  $R$  to make it a transitive relation, is \_\_\_\_.
22. If  $(1 - x^3)^{10} = \sum_{r=0}^{10} a_r x^r (1 - x)^{30-2r}$ , then  $\frac{9a_9}{a_{10}}$  is equal to \_\_\_\_.
23. Let the line  $x - y = 4$  intersect the circle  $C: (x - 4)^2 + (y + 3)^2 = 9$  at the points  $Q$  and  $R$ . If  $P(\alpha, \beta)$  is a point on  $C$  such that  $PQ = PR$ , then  $(6\alpha + 8\beta)^2$  is equal to \_\_\_\_.
24. Let the image of the point  $P(0, -5, 0)$  in the line  $\frac{x-1}{1} = \frac{y+1}{-2} = \frac{z+1}{1}$  be the point  $R$  and the image of the point  $Q(0, \frac{-1}{2}, 0)$  in the line  $\frac{x-1}{-1} = \frac{y+9}{4} = \frac{z+1}{1}$  be the point  $S$ . Then the square of the area of the parallelogram PQRS is \_\_\_\_.
25. Let  $f(x) = \begin{cases} x^3 + 8; & x < 0, \\ x^2 - 4; & x \geq 0, \end{cases}$  and  $g(x) = \begin{cases} (x-8)^{1/3}; & x < 0, \\ (x+4)^{1/2}; & x \geq 0. \end{cases}$   
Then the number of points, where the function  $g \circ f$  is discontinuous, is \_\_\_\_.

### Physics Section A

26. The percentage error in the calculated volume of a sphere, if there is 2% error in its diameter measurement, is \_\_\_\_.
- (A) 1 (B) 2 (C) 6 (D) 8
27. Match List - I with List - II.
- |   | List - I               |      | List - II              |
|---|------------------------|------|------------------------|
| A | Boltzmann constant     | I.   | $[M^{-1} L^3 T^{-2}]$  |
| B | Stefan's constant      | II   | $[ML^2 T^{-1}]$        |
| C | Planck's constant      | III. | $[ML^2 T^{-2} K^{-1}]$ |
| D | Gravitational constant | IV.  | $[ML^0 T^{-3} K^{-4}]$ |
28. A solid sphere (A) of mass  $5m$  and a spherical shell (B) of mass  $m$ , both having same radius, are placed on rough surface. When a force of same magnitude is applied tangentially at the highest points of A and B, they start rolling without slipping with an acceleration of  $a_A$  and  $a_B$ , respectively. The ratio of  $a_A$  and  $a_B$  is \_\_\_\_.
- (A) 5 : 21 (B) 6 : 10 (C) 21 : 25 (D) 1 : 5
29. A body of mass 1 kg moves along a straight line with velocity  $v = 2x^2$ . The work done by the body during displacement from  $x = 0$  to 5 m is \_\_\_\_ J.
- (A) 0 (B) 250 (C) 1250 (D) 1000

Choose the correct answer from the options given below :